

Oleg Belyaev K.

(+34) 630629151
Residence: Cádiz, SP
Nationality: Spain

Oceanographer & GIS Scientist

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Website: www.olegbelyaev.com

Growing up immersed in dual cultures and languages has endowed me with a global perspective and advanced critical thinking skills, enabling me to approach and solve problems effectively. Holding a Bachelor's degree in Biology and a Master's in Coastal Oceanography, my current research focuses on integrating biogeochemical insights with earth observation to understand the complexities of marine ecosystems, particularly in the Antarctic region. My academic journey is driven by a passion for understanding and preserving marine biodiversity, leveraging my expertise in both theoretical and practical aspects of marine science. Outside academia, I enjoy exploring diverse cultures, engaging in outdoor activities, and spending quality time with friends and family.

RESEARCH EXPERIENCE

PHD CANDIDATE / DICHOSO PROJECT

Jan 2024 — Ongoing

Institute of Marine Sciences of Andalusia (ICMAN-CSIC)

Cádiz, Spain

- **Project DICHOSO** (Ref: PID2021-1257830B-I00): Describing biogeochemical features and forecasting future trends of Deception Island, Antarctica.
- Integrating physical-biological interactions into biochemical budgets under PIs Emma Huertas & Antonio Tovar-Sánchez.

REMOTE SENSING SCIENTIST / MARINE STRATEGIES

May 2023 — Dec 2023

Institute of Marine Sciences of Andalusia (ICMAN-CSIC)

Cádiz, Spain

- **Marine Strategies Project** (PI: Gabriel Navarro): Developed satellite-based algorithms for detecting marine heat events and quantifying cumulative impacts on pelagic habitats.
- Continued the technical development of the Andalusian Marine Observatory.

REMOTE SENSING SCIENTIST / EUROSEA PROJECT

Aug 2022

Marine Institute (MI)

Galway, Ireland

- **Project EuroSea** (Ref: 862626): Collaborated on the development of an observation and warning system for extreme marine events (Irish component).

REMOTE SENSING SCIENTIST / SAT4ALGAE

Nov 2021 — Mar 2023

Institute of Marine Sciences of Andalusia (ICMAN-CSIC)

Cádiz, Spain

- **Project Sat4Algae** (Ref: PY20-00244): Generated satellite tools for the detection of macroalgae and microalgae blooms.
- Contributed to the Andalusian Marine Observatory (SOLAUT project) to enhance satellite-based environmental monitoring.

FIELDWORK EXPERIENCE

XXXVII SPANISH ANTARCTIC CAMPAIGN

February 10 — March 26, 2024

"Gabriel de Castilla" Spanish Antarctic Base / "Hespérides" Research Vessel

Deception Island, Antarctica

- 22 days of fieldwork at the "Gabriel de Castilla" base at Deception Island.
- 12 days onboard the R/V "Hespérides".
- Participation as part of the DICHOSO Antarctic Project Framework.

PUBLICATIONS & DATASETS

- Belyaev, O., Román A., Belliure J., Navarro G., Barbero L., & Tovar-Sánchez A. *Assessing Topographic Features and Population Abundance in an Antarctic penguin colony through UAV-Based Deep-Learning Models*. *International Journal of Applied Earth Observation and Geoinformation*, 2024.
- Belyaev, O., Huertas, I.E., Navarro G., Amaya-Vías S., De La Paz M., Sparaventi E., Heredia S., Sukekava C. F., Laglera L. M., & Tovar-Sánchez A. *Hydrothermal alteration of seawater biogeochemistry in Deception Island (South Shetland Islands, Antarctica)*. *Frontiers in Marine Science*, 2024.
- Fernández-Barba, M., Belyaev, O., Huertas, I.E., & Navarro G. *Marine heatwaves in a shifting Southern Ocean induce dynamical changes in primary production*. *Communications Earth & Environment*, 2024.
- Fernández-Barba, M., Belyaev, O., Huertas, I. E., & Navarro, G. *Marine Heatwaves in a Changing Southern Ocean: Heat Budget Analysis in Modular Ocean Model v4p1 (ESM2M GFDL) and Causal Inference through Convergent Cross Mapping*. *DIGITAL.CSIC.*, 2024. [DATASET]
- Pereiro D., Belyaev, O., ... & Cusack C. *An observational and warning system for the aquaculture sector*. *Frontiers in Marine Science*, 2024.

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- **Belyaev, O.**, Sparaventi, E., Navarro G., Rodríguez-Romero A., & Tovar-Sánchez A. *The contribution of penguin guano to the Southern Ocean iron pool*. *Nature Communications*, 2023.
- **Belyaev, O.**, Sparaventi, E., Navarro G., Rodríguez-Romero A., & Tovar-Sánchez A. *Training dataset for object detection - Penguins from UAV, Ver. 1*, Australian Antarctic Data Centre, 2023. [DATASET]
- Pereiro D., Cusack C., Dunbar M. B., Navarro G., Alvarez-Berastegui, D., O'Carroll, J., **Belyaev, O.**, ... & Wilkes, R. *Connections between "Extreme Marine Events" and Biological EOVS Report*. *GEOMAR OceanRep*, 2022.
- **Belyaev, O.**, Caballero, I., & Navarro G. *Servicio de monitorización oceánica en tiempo cuasi real a través de la teledetección: Observatorio Marino de Andalucía*. *Conference Book of the Spanish Remote Sensing Association*, 2021. ISBN: 978-84-9769-383-7
- **Belyaev, O.**, Román A., Navarro G., & Tovar-Sánchez A. *Penguin population quantification from UAV imagery using deep-learning in Deception Island (South Shetland Islands, Antarctica)*. *Conference Book of the Young Sea Researchers Conference (JIs del Mar)*, 2021. ISBN: 978-84-09-36126-7

CONFERENCES

- **Living Planet Symposium 2025 (LPS2025)**, The European Space Agency. International congress, Vienna, Austria, June 2025. *Poster presentation* – Evaluating topographic characteristics and population density in an Antarctic penguin colony using UAV-driven deep learning models.
- **XX Congress of the Spanish Association of Remote Sensing (AET) 2024**. National congress, Cádiz, June 2024. Listener & co-organizer.
- **II International Meeting on Knowledge and Blue Economy – InnovAzul 2022**. Meeting of students with companies and professionals from the Blue Economy sectors, Cádiz, November 29 - December 2, 2022. Listener.
- **2022 Sustainable Ocean Summit (SOS) & Global Blue Finance Summit (BlueFIN)**. International congress, Barcelona, October 2022. Listener.
- **VIII International Symposium on Marine Sciences (ISMS) 2022**. International congress, Gran Canaria, July 2022. *Poster presentation* – Satellite monitoring of forecasted oceanic variables through the Marine Observatory of Andalusia.
- **XIX Congress of the Spanish Association of Remote Sensing (AET) 2022**. National congress, Pamplona, June-July 2022. *Poster presentation* – Ocean monitoring service through remote sensing: Andalusian Marine Observatory.
- **EuroSea Annual Meeting 2022**. European congress, Cádiz, May 2022. *Oral presentation* - Development of a near-real-time remote sensing ocean monitoring service: the Andalusian Marine Observatory.
- **III Congress of Young Marine Researchers (JIS)**. International congress, Motril, Granada, September 2021. *Oral presentation* – Quantification of penguin populations using neural networks on Deception Island (South Shetland Islands, Antarctica).
- **XXXIII Conference of the Spanish Association of Entomology**. National congress, Almería, November 15-18, 2017. *Oral presentation* - Bergmann's Temporal Rule? Contrasting effects of temperature and solar radiation on the body size of dung beetles across circadian rhythms.
- **IX Spanish Space Students Conference at LEEM-UPM**. Madrid, 2017. *Co-organizer* & Listener.

PROJECTS

- **DICHOSO - Contribution of water masses of Deception Island to biogeochemical inventories of the Southern Ocean: current budgets and future trends (Ref: PID2021-125783OB-I00)**.
PI: Emma Huertas & Antonio Tovar-Sánchez. *PhD Program*. 2024 — ongoing
- **Marine Strategies - Application of models for the quantification of cumulative impacts on pelagic habitats in pilot areas**.
PI: Gabriel Navarro *Development of satellite-based algorithms for the detection of marine heat events*. 2023

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- **EuroSea - Improving the European ocean observing and forecasting system in a global context. (Ref: 862626).**
Development of Irish part of the Marine Observatory within the framework of this project.. 2022
- **Sat4Algae Project (Ref: PY20-00244. 2021-2022), Project of Excellence, Junta de Andalucía. PI: Gabriel Navarro.**
Development of the Andalusian Marine Observatory within the framework of this project.. 2021 — 2023
- **PiMetAn Project (Ref: RTI2018-098048-B-100), Ministry of Science and Innovation. PI: Antonio Tovar-Sánchez.**
Master's thesis developed within the framework of this project. 2021

EDUCATION

Ph.D. in Marine Science and Technology, CADIZ UNIVERSITY, Spain 2024 — ongoing
Antarctic Biogeochemistry and Remote Sensing research
MSc. in Oceanography, CADIZ UNIVERSITY, Spain 2020 — 2021
Deep understanding of biological, physical, chemical and geological processes in the coastal oceanic systems.
BSc. in Molecular Biology (Erasmus+ Program), UMEÅ UNIVERSITY, Sweden 2018 — 2019
Advanced molecular biology and genetics laboratory techniques, Neurobiology.
BSc. in Biology, ALCALA UNIVERSITY, Spain 2015 — 2018
Understanding of the basic structures and dynamics of living organisms and its interactions with their environment.

TECHNICAL PROFICIENCIES

Text Edition and Database	Git, LaTeX
GIS Software	QGIS, UGCS, Google Earth Engine
Scientific Programming	Python, JavaScript (for GEE), MATLAB
Design	Adobe Illustrator, Photoshop, AfterEffects

COURSES

Expert in Data Science Datamecum
PhD Course on spatial data analysis in ecology with R Cádiz University
A1-A3 & A2 UAV Certified Operator AESA
SSI Divemaster +220 logged dives and +100 OW Assisted OW Courses, Nitrox Certified. Princess Divers
Google Earth Engine CSIC
Field-work Risk Management CSIC

AREAS OF EXPERTISE

- Geographical Information Systems, UAVs, Coastal Oceanography Processes
- Neural-network based object detection
- Scientific illustration
- Teamwork, communication, pathfinder
- Cross-Functional Team Leadership

LANGUAGES

Spanish, Russian	Native level
English	Fluent speaker
French	Basic speaker

PERSONAL ACTIVITIES AND ADDITIONAL INFORMATION

Coral Restoration program with OceanQuest and Sea Sepherd in Thailand	2020
European Driving License B	2016
Co-Manager of Literature and Philosophy Club, <i>La flauta Magica</i> in Madrid	2015-2017

PRIVACY

I hereby authorise the use of my personal data in accordance to the GDPR 679/16-"European regulation on the protection of personal data"